MIL-P-43116A
28 SEPTEMBER 1964

SUPERSEDING
MIL-P-43116(GL)
23 NOVEMBER 1962
MIL-P-12415(CE)
6 JANUARY 1953
MIL-P-19827 (DOCKS)
7 FEBRUARY 1957

MILITARY SPECIFICATION

PIKE, WIRE AND PIKES, POLE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

* 1.1 *Scope.* This specification covers wire and pole pikes for use by linemen and fire fighters (see 6.1).

\# 1.2 **Classification.** Wire and pole pikes shall be of the following types, styles and sizes as specified (see 6.2):

Type I	— For wire or cable use.
Type II	— For utility pole erection.
Style A	— Without safety guard.
Style B	— With safety guard.
Size 1	— 8 foot.
Size 2	— 10 foot.
Size 3	— 14 foot.
Type III	— For fire fighting.
Style A	— Crash fire fighting.
Style B	— Structural fire fighting.
Size 1	— 8 foot.
Size 2	— 16 foot.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-367	— Aluminum-Alloy Forgings, Heat Treated.
QQ-I-666	— Iron Castings, Malleable.
QQ-P-416	— Plating, Cadmium (Electrodeposited).
QQ-S-633	— Steel Bars, Carbon, Cold Finished and Hot Rolled (General Purpose).
QQ-S-698	— Steel, Sheet and Strip, Low-Carbon.
QQ-Z-325	— Zinc Coating, Electrodeposited, Requirements For.

FSC 5120

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WU-T-789 — Tube, Aluminum Alloy, Round, Square, Rectangular, and Other Shapes, Drawn, Seamless, 6061 and 6062.

WW-T-825 — Tube, Magnesium-Alloy, Extruded.

MILITARY

MIL-M-3171 — Magnesium Alloy, Processes For Corrosion Protection Of.

MIL-H-13048 — Steel Forgings (For General Purpose Application).

MIL-H-15424 — Hand Tools; Packaging Of.

MIL-P-16610 — Pins, Taper, Carbon Steel.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MS 9048 — Pin-Spring, Steel, Cadmium Plated.

MS 171401 — Pin-Spring, Corrosion-Resistant Steel.

* DRAWINGS

ARMY NATICK LABORATORIES

12-1-60 — Pike, Wire, Type I, For Wire or Cable Use; Assembly and Details.

12-1-153 — Pike Pole, Type II, Style "A" and "B"

12-1-154 — Pike Pole, Type III, Style "A" and "B"

(Miniature copies of Drawings 12-1-60, 12-1-153 and 12-1-154, identified as figures 1, 2, and 3, are attached for information purposes only.)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Reproduction sample. When specified (see 6.2), before production is commenced, a sample of the finished item shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

* **3.2 Materials.** The materials specified shall conform to the referenced specifications and requirements specified herein. Materials not definitely specified shall be of commercial quality, suitable for the purpose intended.

* 3.2.1 *Wood.* Wood shall be any of the following species: Douglas Fir, White Ash, Hickory, Yellow Birch, Hard Maple, Beech, Rock Elm, Oak or Pecan. The wood shall be seasoned to not more than 18 percent moisture content at the time of fabrication when tested as specified in 4.3.1.1. The wood shall have not more than 12 growth rings per inch, except that Fir shall have not less than 8 rings nor more than 22 rings, Hickory shall have not more than 22 rings, and Ash not more than 17 rings per inch. The slope of grain shall not exceed 1 in 15, with defects limited to small burls, tight knots under ¼ inch in diameter, and slight streaks or stain. The number of permissible defects

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shall be two (2) for each 18 inches or fraction thereof, with defects at least 6 inches apart.

3.2.2 *Bar, carbon steel.* The carbon steel bars shall be C1030 to C1040, cold finished, conforming to QQ-S-633.

3.2.3 *Sheet, steel.* The steel sheet shall be 1015 to 1020, cold rolled, drawing quality, special killed, conforming to QQ-S-698.

3.2.4 *Casting, malleable iron.* The malleable iron casting shall be grade I or II conforming to QQ-I-666.

3.2.5 *Forgings, steel.* The steel forgings, shall be composition C1030 to C1040 and shall conform to the requirements of MIL-S-13048.

3.2.6 *Forgings, aluminum alloy.* Aluminum alloy forgings shall be composition 2014 or 2017 conforming to QQ-A-367.

3.2.7 *Tubing, aluminum alloy.* The aluminum tube shall be composition 6061, temper T6, type I, conforming to WW-T-789.

3.2.8 *Tubing, magnesium alloy, extruded.* The magnesium alloy tube shall be composition AZ61A conforming to WW-T-825.

3.2.9 *Pipe, steel.* When used, the steel pipe shall be of the type normally furnished in commercial practice.

3.2.10 *Pin, taper, steel.* The steel taper pin shall conform to MIL-P-16610.

3.2.11 *Pin, spring, steel.* The steel spring pins shall conform to MS 9048.

3.2.12 *Pin, spring, corrosion-resistant steel.* The corrosion-resistant steel spring pin shall conform to MS 171401.

3.2.13 *Bearing, nylon.* The nylon bearing shall be of the type normally furnished in commercial practice.

3.2.14 *Coating, zinc.* Zinc coating shall conform to type II, class 2 of QQ-Z-325.

3.2.15 *Plating, cadmium.* Cadmium plating shall conform to class 1, type II of QQ-P-416.

3.3 Construction.

3.3.1 *Type I, wire pike.* The type I wire pike shall be fabricated of materials specified in 3.2 as applicable and in accordance with details and dimensions shown on Drawing No. 12-1-60.

3.3.2 *Type II, style A and B pike poles.* The type II, style A and B pike poles shall conform to Drawing No. 12-1-153 and be fabricated of the materials specified herein.

3.3.2.1 *Heads.* Pike pole heads shall be one piece, steel forgings specified in 3.2.5, and shall have a spike shaped point not less than 4 inches long, with a shoulder to bear against the end of the handle and a tang of suitable length to be embedded in the handle.

3.3.2.2 *Ferrule.* The ferrule shall be fabricated of drawn steel specified in 3.2.3. When assembled, a rivet shall pass completely through the ferrule, the handle, and tang of the head for secure attachment of components.

3.3.2.3 *Guard.* The guard for style B pike poles shall be fabricated of steel specified in 3.2.3 or 3.2.5. The guard shall be riveted to the ferrule and provisions shall be made for locking the guard in an opened or closed position.

3.3.2.4 *Handles.* The handles shall be fabricated of the wood specified in 3.2.1, 2 inches in diameter, and be of the strength and rigidity normally furnished for pole erection use.

3.3.3 *Type III, style A and B pike pole.* The type III, style A and B pike poles shall

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conform to Drawing No. 12-1-154 and be fabricated of the materials specified herein.

3.3.3.1 *Heads.* The head for style A pike pole shall be of steel forging specified in 3.2.5 The head for style B pike pole shall be fabricated of materials specified in 3.2.4, 3.2.5 or 3.2.6.

3.3.3.2 *Handles.* Unless otherwise specified (see 6.2), handles shall be fabricated of wood specified in 3.2.1. When aluminum alloy tubing or magnesium alloy tubing is specified, the material shall conform to the requirements specified in 3.2.7 or 3.2.8 respectively. When metal handles are used, the end opposite the heads shall be closed. Magnesium alloy handles shall be insulated from steel and iron parts to prevent electrolytic action and the insulation shall be suitable for use under conditions normally encountered in fire fighting.

3.4 **Assembly.** When completely assembled, pike pole heads shall not become loose nor pull free of the handles when tested as specified in 4.4.1 and 4.4.2.

3.5 **Finish.** All metal parts, except the corrosion-resistant steel pins, wood, aluminum and magnesium alloy handles, shall be zinc coated or cadmium plated as specified in 3.2.14 or 3.2.15. Wood handles shall be smoothly finished by sanding and waxing only. Aluminum handles shall be finished and painted in accordance with the manufacturer's standard practice. Magnesium alloy handles shall be treated in accordance with type III or MIL-M-3171.

3.6 **Marking for identification.** The manufacturer's name or trade mark of such known character that the source of manufacture may be readily determined shall be legibly embossed or impressed on the handle and head. Marking shall be in accordance with MIL-STD-130.

3.7 **Workmanship.** The finished wire and pole pike shall conform to the quality and grade of product established by this specification. There shall be no evidence of burrs, nicks, rough edges, tacky finish, chipping or rust.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilites or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 **Preproduction sample.** When a preproduction sample is required, it shall be examined for defects in table I, dimensions specified, and tested as specified in 4.4.1 and 4.4.2.

4.3 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shal be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3.1 *Inspection of materials and components.* In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.3.1.1 *Testing of wood.* Tests shall be performed on each lot of lumber offered for inspection at time of fabrication for moisture content in accordance with MIL-W-6110 for compliance with 3.2.1. Three determinations shall be taken on each sample unit and results shall be the average of

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the three determinations. The inspection level shall be S–4 with an acceptable quality level (AQL) of 2.5, expressed in terms of defects per hundred units. The sample unit shall be one piece of lumber.

4.3.2 *In-process examination.* Examination shall be made of the following operations to establish that no deviation is made from the indicated requirements. Whenever a deviation is noted, correction shall be made to the affected items and lot in process.

Insulation	—Magnesium alloy handles to be insulated from steel and iron parts, (see 3.3.—3.2).
Finish	—Wood handles to be smoothly finished by sanding and waxing and magnesium alloy handles treated as specified, (see 3.5).

4.3.3 *End item inspection.* The lot shall consist of all wire and pole pikes of one type, style and size submitted for acceptance at one time. The sample unit shall be one (1) completely fabricated wire or pole pike.

* 4.3.3.1 *Visual examination.* Visual examination of the end item shall be in accordance with the classification of defects in table I. The inspection level shall be level II, with an AQL of 2.5 for major defects and an AQL of 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. *Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship.	Any sharp burr, nicks or rough edges	X	
	Any part or component missing or not specified type.	X	
	Any part or component fractured, split, deformed, broken or otherwise impaired.	X	
	Misplaced, loose, not in proper alignment, not properly assembled or secured.	X	
	Evidence of rust or corrosion	X	
Wood handles	Number of growth rings not as specified	X	
	Silver or splinter	X	
	Slope of grain not as specified	X	
	Small burls, tight knots under ¼ inch diameter, slight streaks or stain (Note: The number of admissible defects shall be two for each 18 inches of handle length or fraction thereof, with defects at least 6 inches apart).		X
Metal handles	End opposite head not closed		X
Finish	Not as specified		X
Identification marking	Missing, incomplete, not legible, not permanent or not as specified.		X

* 4.3.3.2 *Dimensional examination.* Examination shall be made for compliance with dimensional requirements specified herein. Any deviation therefrom shall constitute a defect. The inspection level shall be S–2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.3.3 *Testing of end item.* Testing shall be performed on each sample unit in accordance with 4.4.1 and 4.4.2 to determine compliance with the requirements of 3.4. The sample unit shall be one completely fabricated wire or pole pike. The inspection level shall be S–4 with an AQL of 2.5, expressed

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in terms of defects per hundred units.

* 4.3.4 *Examination of preparation for delivery.* Examination shall be made to determine compliance with preservation and packaging, packing and marking requirements in accordance with MIL-H-15424.

4.4 **Tests.**

4.4.1 *Pull test for type II pole pike.* The type II pole pike head shall be securely clamped in a vise. A load of 100 pounds shall be applied to the handle in the opposite direction of the head for 5 minutes. The pole pike shall be examined to determine compliance with the requirements of 3.4.

4.4.2 *Load test for type III pole pike.* The type III pole pike handle shall be securely clamped in a vertical position with the head down, and a load of not less than 400 pounds applied to the hook of head in a downward direction for 5 minutes. The pole pike shall be examined to determine compliance with the requirements of 3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing shall be in accordance with the applicable level A, B or C requirements of MIL-H-15424 for pole, pike (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type I wire pike covered by this specification is intended for use by linemen to position or reclaim light weight wire and cables in the field. The type II pole pike covered by this specification is intended for use by linemen in the erection of utility poles. The type III pole pike covered by this specification is intended for use by fire fighters in crash fire fighting and structural fire fighting.

* **6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, style and size required (see 1.2).
(c) When preproduction sample is required (see 3.1).
(d) Specify when other than wood handle is required (see 3.3.3.2).
(e) Selection of applicable levels of preservation and packaging and packing (see 5.1).

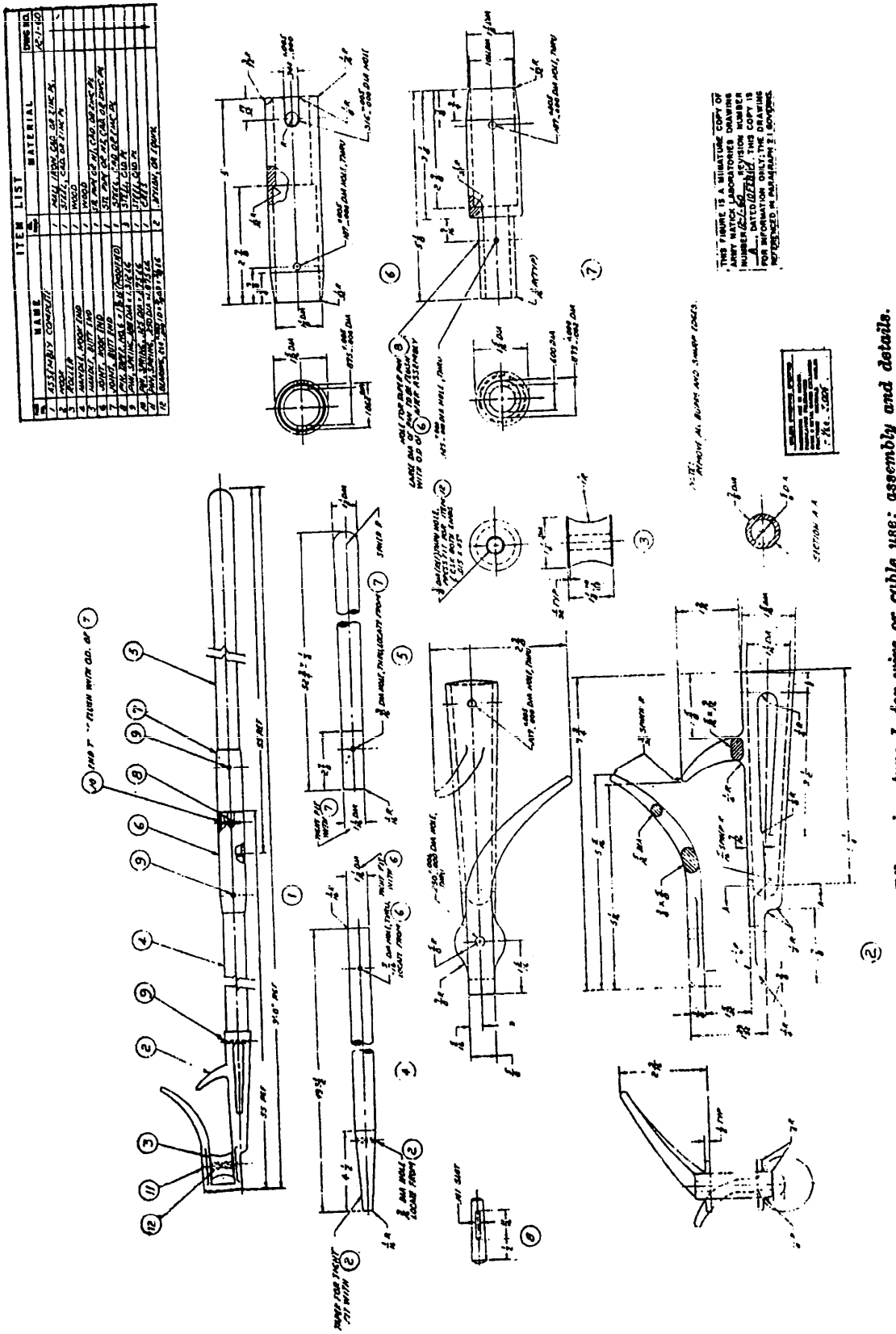
6.3 The outside margins of this document have been marked to indicate where changes from the previous issue have been made. An asterisk (*) indicates a substantive change in the paragraph text. A double asterisk (**) indicates deletion of a paragraph bearing the same number. The sign (#) indicates the addition of a completely new paragraph. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

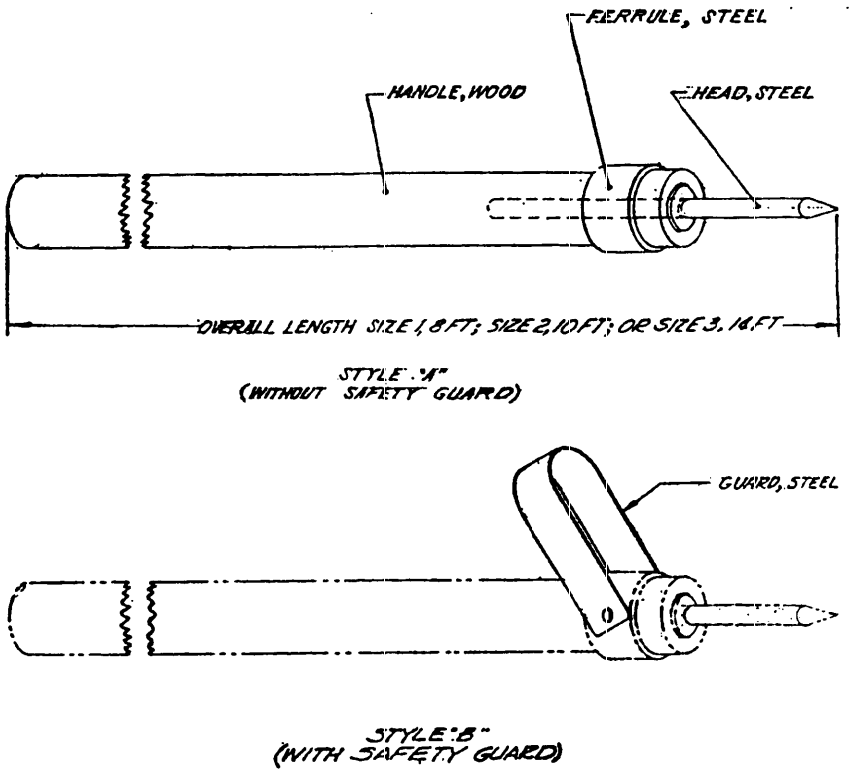
Custodians:
Army—GL
Navy—YD
Air Force—69
Interested activities:
Review:
Army—MO
Navy—YD
Air Force—69
User:
Army—EL, SM
Navy—MC

Preparing activity:
Army—GL
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ITEM LIST
NAME
MATERIAL
DWG NO
ASSEMBLY COMPLETE
BODY
WASHER, NOSE END
WASHER, BUTT END
JOINT, BOX END
JOINT, PIN END
PIN, BOX END
PIN, PIN END
SPRING
SPRING
SPRING



NOTE:

THESE DESIGNS ARE ILLUSTRATIVE ONLY & ARE NOT INTENDED TO BE RESTRICTIVE.

FIGURE 2. *Pike pole, type II, style A and B.*

NEXT ASSY	USED ON
APPLICATION	
DO NOT	APPLY PART NO.
DO	AS SPECIFIED

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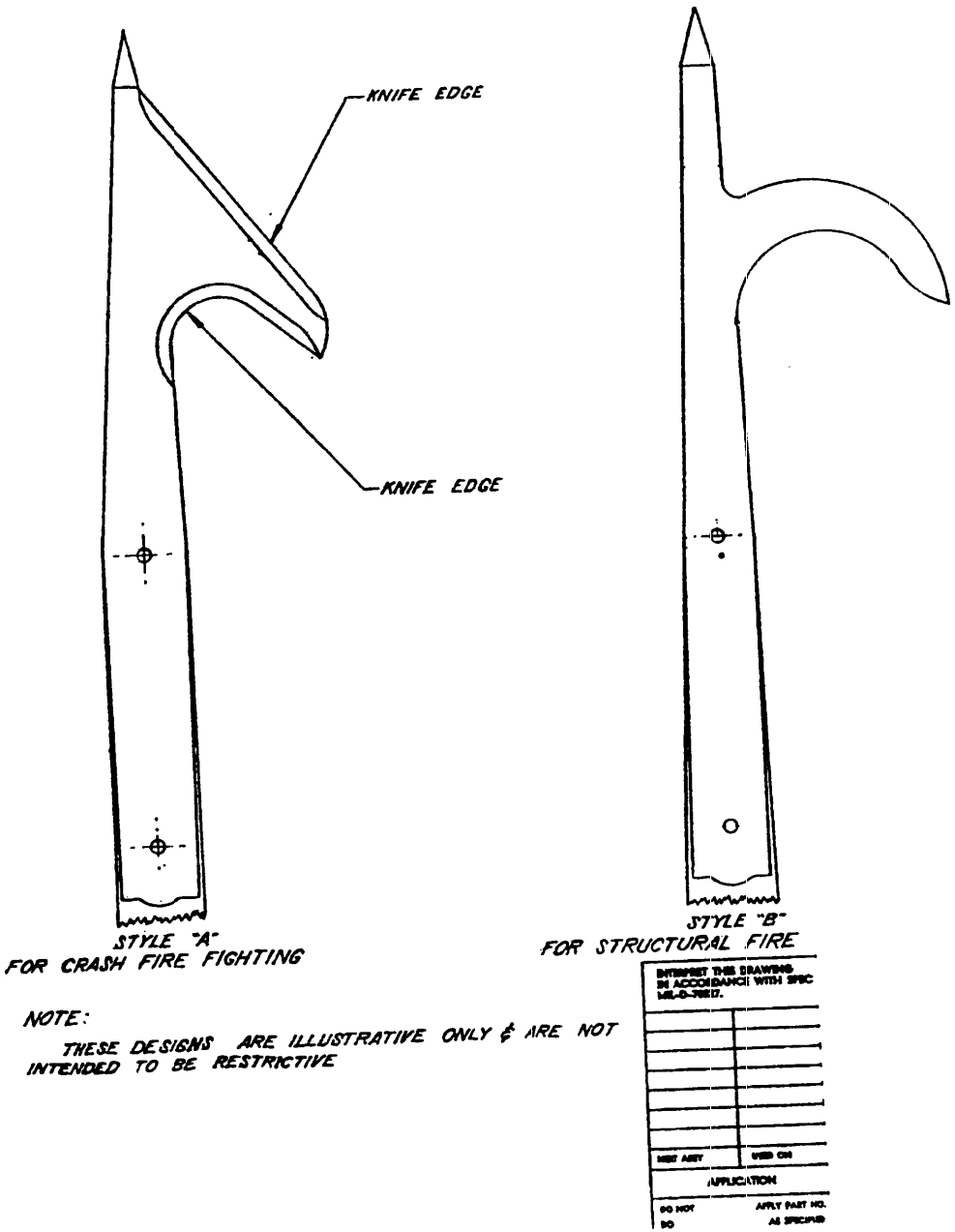


FIGURE 3. *Pike pole, type III, style A and B.*

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